

MAT110

Lecture 07

Relative Maximum, Relative Minimum, and Curve Sketching

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Main focus: First derivative test, second derivative test, higher derivative test, and a systematic guideline for sketching graphs of functions.

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1 Relative Maximum and Relative Minimum

Definition: Relative Maximum

Let f be defined on an interval containing x_0 . The function f has a **relative maximum** at $x = x_0$ if there exists a small open interval around x_0 such that

$$f(x) \leq f(x_0)$$

for every x in that interval and in the domain of f .

Definition: Relative Minimum

Let f be defined on an interval containing x_0 . The function f has a **relative minimum** at $x = x_0$ if there exists a small open interval around x_0 such that

$$f(x) \geq f(x_0)$$

for every x in that interval and in the domain of f .

Geometrical Meaning

A relative maximum point is a point where the graph changes direction from increasing to decreasing. A relative minimum point is a point where the graph changes direction from decreasing to increasing.

2 Critical Points

Critical Point

A point $x = x_0$ in the domain of f is called a **critical point** if

$$f'(x_0) = 0$$

or if $f'(x_0)$ does not exist.

Important Note

Every relative maximum or relative minimum can occur at a critical point. So, to search for relative extrema, first find the points where $f'(x) = 0$ or where $f'(x)$ does not exist.

3 First Derivative Test

First Derivative Test

Let f be continuous at a critical point x_0 and differentiable on the left and right of x_0 .

- (a) If $f'(x) > 0$ on an open interval to the left of x_0 and $f'(x) < 0$ on an open

interval to the right of x_0 , then f has a **relative maximum** at x_0 .

(b) If $f'(x) < 0$ on an open interval to the left of x_0 and $f'(x) > 0$ on an open interval to the right of x_0 , then f has a **relative minimum** at x_0 .

(c) If $f'(x)$ has the same sign on both sides of x_0 , then f has neither a relative maximum nor a relative minimum at x_0 .

Sign of f' before x_0	Sign of f' after x_0	Conclusion
+	−	Relative maximum
−	+	Relative minimum
+	+	Neither maximum nor minimum
−	−	Neither maximum nor minimum

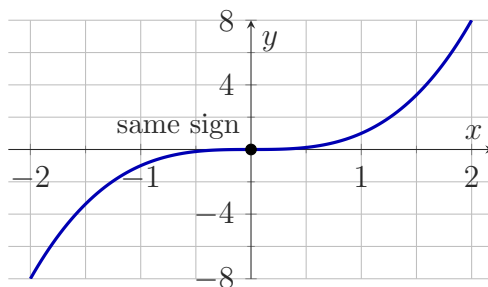
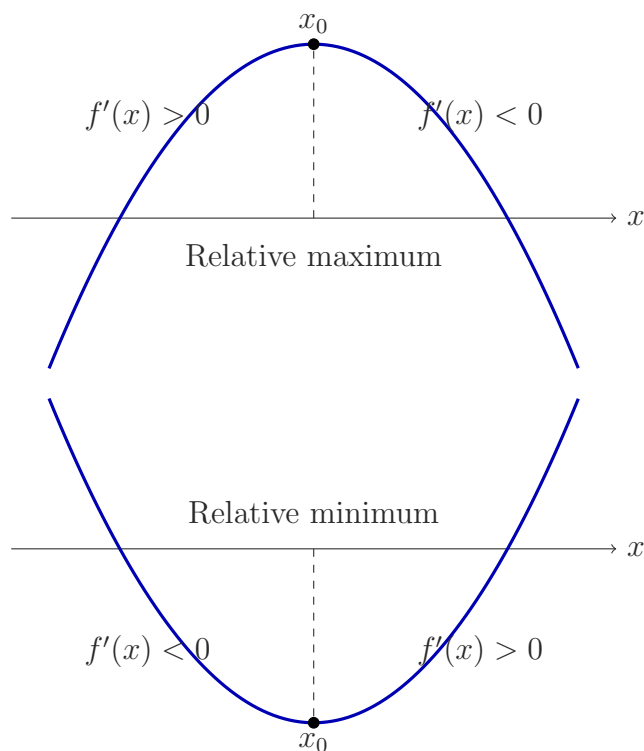


Figure: For $y = x^3$, $f'(x) = 3x^2 \geq 0$ on both sides of 0, so there is no relative extremum at 0.

4 Example Using the First Derivative Test

Example

The function

$$f(x) = 3x^{5/3} - 15x^{2/3}$$

has a relative maximum at $x = 0$ and a relative minimum at $x = 2$. Confirm this using the first derivative test.

Solution

Given

$$f(x) = 3x^{5/3} - 15x^{2/3}.$$

Differentiating with respect to x , we get

$$\begin{aligned} f'(x) &= 3 \cdot \frac{5}{3}x^{2/3} - 15 \cdot \frac{2}{3}x^{-1/3} \\ &= 5x^{2/3} - 10x^{-1/3} \\ &= \frac{5x - 10}{x^{1/3}} \\ &= \frac{5(x - 2)}{x^{1/3}}. \end{aligned}$$

Critical points occur when $f'(x) = 0$ or $f'(x)$ does not exist.

$$f'(x) = 0 \Rightarrow \frac{5(x - 2)}{x^{1/3}} = 0 \Rightarrow x = 2.$$

Also, $f'(x)$ does not exist at $x = 0$, but $f(0)$ exists. Therefore the critical points are

$$x = 0, \quad x = 2.$$

Now we examine the sign of $f'(x)$.

Interval	Sign of $f'(x) = \frac{5(x - 2)}{x^{1/3}}$	Behavior of f
$(-\infty, 0)$	+	Increasing
$(0, 2)$	-	Decreasing
$(2, \infty)$	+	Increasing

Since the sign of f' changes from $+$ to $-$ at $x = 0$, the function has a relative maximum at $x = 0$.

Since the sign of f' changes from $-$ to $+$ at $x = 2$, the function has a relative minimum at $x = 2$.

The corresponding function values are

$$f(0) = 0,$$

and

$$f(2) = 3 \cdot 2^{5/3} - 15 \cdot 2^{2/3} = -9 \cdot 2^{2/3} = -9\sqrt[3]{4}.$$

Hence the relative maximum point is

$$(0, 0),$$

and the relative minimum point is

$$(2, -9\sqrt[3]{4}).$$

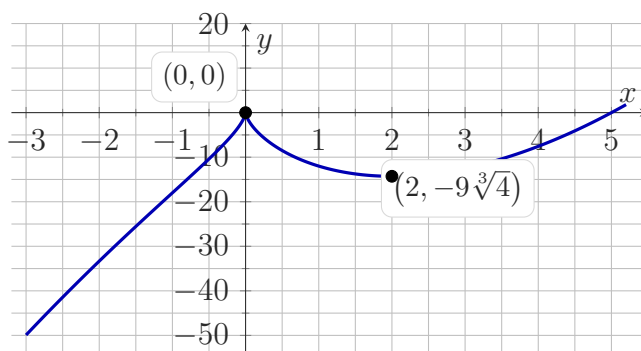


Figure: Graph of $f(x) = 3x^{5/3} - 15x^{2/3}$.

5 Second Derivative Test

Motivation

The first derivative test uses the sign of $f'(x)$ before and after a critical point. The second derivative test uses the value of $f''(x)$ at a stationary point.

Second Derivative Test

Let f be twice differentiable near x_0 , and suppose that

$$f'(x_0) = 0.$$

Then:

- (a) If $f''(x_0) > 0$, then f has a **relative minimum** at x_0 .
- (b) If $f''(x_0) < 0$, then f has a **relative maximum** at x_0 .
- (c) If $f''(x_0) = 0$, then the test is **inconclusive**. That means f may have a relative maximum, a relative minimum, or neither at x_0 .

Condition	Conclusion
$f'(x_0) = 0$ and $f''(x_0) > 0$	Relative minimum at x_0
$f'(x_0) = 0$ and $f''(x_0) < 0$	Relative maximum at x_0
$f'(x_0) = 0$ and $f''(x_0) = 0$	Test is inconclusive

How to Remember

If $f''(x_0) > 0$, the graph is concave upward near x_0 , so the point behaves like the bottom of a bowl. Hence it gives a relative minimum.

If $f''(x_0) < 0$, the graph is concave downward near x_0 , so the point behaves like the top of a hill. Hence it gives a relative maximum.

6 Example Using the Second Derivative Test

Example

Find the relative extrema of

$$f(x) = 3x^5 - 5x^3.$$

Solution

Given

$$f(x) = 3x^5 - 5x^3.$$

First derivative:

$$\begin{aligned} f'(x) &= 15x^4 - 15x^2 \\ &= 15x^2(x^2 - 1). \end{aligned}$$

For stationary points, put $f'(x) = 0$.

$$\begin{aligned} 15x^2(x^2 - 1) &= 0 \\ x^2(x - 1)(x + 1) &= 0. \end{aligned}$$

Therefore

$$x = 0, \quad x = -1, \quad x = 1.$$

Second derivative:

$$\begin{aligned} f''(x) &= 60x^3 - 30x \\ &= 30x(2x^2 - 1). \end{aligned}$$

Now evaluate $f''(x)$ at the stationary points.

Stationary point	$f''(x)$	Conclusion
$x = -1$	$f''(-1) = -30 < 0$	Relative maximum
$x = 0$	$f''(0) = 0$	Inconclusive
$x = 1$	$f''(1) = 30 > 0$	Relative minimum

Therefore, by the second derivative test, f has a relative maximum at $x = -1$ and a relative minimum at $x = 1$. The point $x = 0$ needs further checking because the test is inconclusive there.

The corresponding function values are

$$f(-1) = 3(-1)^5 - 5(-1)^3 = 2,$$

$$f(1) = 3(1)^5 - 5(1)^3 = -2,$$

and

$$f(0) = 0.$$

Hence the relative maximum point is

$$(-1, 2),$$

and the relative minimum point is

$$(1, -2).$$

Checking the Inconclusive Point x

Since $f''(0) = 0$, the second derivative test gives no conclusion at $x = 0$. We use the first derivative sign around 0.

Since

$$f'(x) = 15x^2(x^2 - 1),$$

we see that $f'(x) < 0$ on both intervals $(-1, 0)$ and $(0, 1)$. Therefore there is no sign change in f' at $x = 0$.

So there is neither a relative maximum nor a relative minimum at $x = 0$.

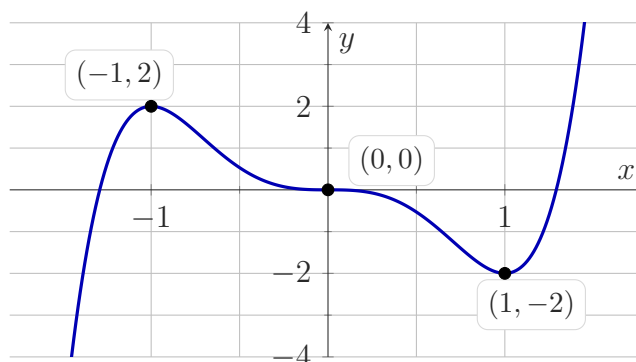


Figure: Graph of $f(x) = 3x^5 - 5x^3$.

7 Higher Derivative Test

Higher Derivative Test

Suppose that f is sufficiently differentiable near x_0 and

$$f'(x_0) = f''(x_0) = \cdots = f^{(k-1)}(x_0) = 0,$$

but

$$f^{(k)}(x_0) \neq 0,$$

where $k \geq 2$.

Then:

- (a) If k is even and $f^{(k)}(x_0) > 0$, then f has a relative minimum at x_0 .
- (b) If k is even and $f^{(k)}(x_0) < 0$, then f has a relative maximum at x_0 .
- (c) If k is odd, then f has neither a relative maximum nor a relative minimum at x_0 .

Connection with Familiar Graphs

For $f(x) = x^4$, the first nonzero derivative at 0 is the fourth derivative, and it is positive. So $x = 0$ gives a relative minimum.

For $f(x) = -x^4$, the first nonzero derivative at 0 is the fourth derivative, and it is negative. So $x = 0$ gives a relative maximum.

For $f(x) = x^3$, the first nonzero derivative at 0 is the third derivative. Since 3 is odd, there is neither a relative maximum nor a relative minimum at 0.

8 General Guidelines for Sketching the Graph of a Function

Curve Sketching Checklist

To sketch the graph of a function, collect the following information step by step.

1. **Domain:** Find all values of x for which the function is defined.
2. **Intercepts:** Find the x -intercepts and y -intercept, if they exist.
3. **Symmetry Test:** Check whether the function is even, odd, or neither.
4. **Asymptotes:** Find vertical, horizontal, or oblique asymptotes, if any.
5. **Intervals of Increase and Decrease:** Use the sign of $f'(x)$.
6. **Local Maximum and Minimum:** Use the first derivative test or the second derivative test.

7. **Concavity and Points of Inflection:** Use the sign of $f''(x)$.
8. **Sketch the Curve:** Combine all information and draw the graph carefully.

Symmetry Tests

Let f be a function.

- (a) If $f(-x) = f(x)$, then f is an even function. Its graph is symmetric about the y -axis.
- (b) If $f(-x) = -f(x)$, then f is an odd function. Its graph is symmetric about the origin.

Concavity and Inflection

Let f be twice differentiable on an interval I .

- (a) If $f''(x) > 0$ on I , then the graph of f is concave upward on I .
- (b) If $f''(x) < 0$ on I , then the graph of f is concave downward on I .
- (c) A point where the concavity changes is called a point of inflection.

9 Summary

Key Takeaways

1. A relative maximum occurs when the function changes from increasing to decreasing.
2. A relative minimum occurs when the function changes from decreasing to increasing.
3. The first derivative test checks the sign change of $f'(x)$.
4. The second derivative test checks the value of $f''(x)$ at a stationary point.
5. If $f''(x_0) = 0$, then the second derivative test is inconclusive.
6. Curve sketching requires domain, intercepts, symmetry, asymptotes, increasing/decreasing intervals, extrema, concavity, and inflection points.

10 Practice Problems

Try Yourself

1. Find the relative maximum and relative minimum of $f(x) = x^3 - 3x^2 + 2$ using the first derivative test.
2. Find the relative extrema of $f(x) = x^4 - 4x^2$ using the second derivative test.
3. Determine whether $x = 0$ gives a relative maximum, relative minimum, or neither for $f(x) = x^5$.
4. Sketch the graph of $f(x) = x^3 - 3x$ by finding its critical points and intervals of increase and decrease.
5. Use the curve sketching checklist to sketch $f(x) = \frac{x}{x^2 + 1}$.

11 Practice Problems

1. Find the relative extrema (maxima/ minima) using both the first and second derivative tests.

$$(i) f(x) = 2x^3 - 9x^2 + 12x \quad (ii) f(x) = \frac{x}{2} - \sin x, \quad 0 < x < 2\pi.$$

2. Use the given derivative to find all critical numbers of f and determine whether a relative maximum, relative minimum, or neither occurs there.

$$(i) f'(x) = x^3(x^2 - 5) \quad (ii) f'(x) = \frac{x^2 - 1}{x^2 + 1}.$$